

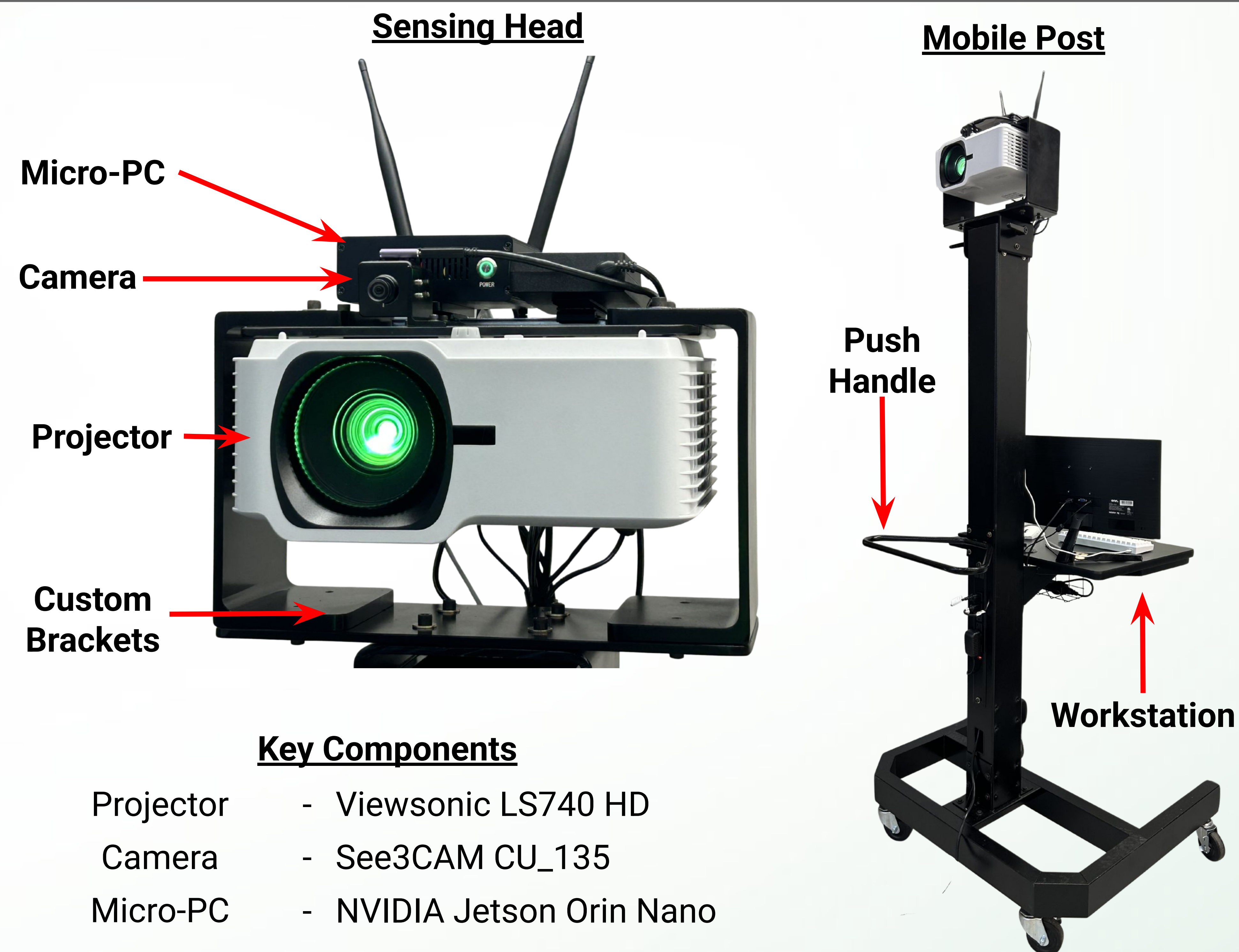
Objective

Design an *automatic projection system* that uses *computer vision* to digitally project an arrangement of decals onto truck doors, removing the need for manual decal measurement and ensuring *fast, accurate, and consistent placement*.

Capabilities & Features

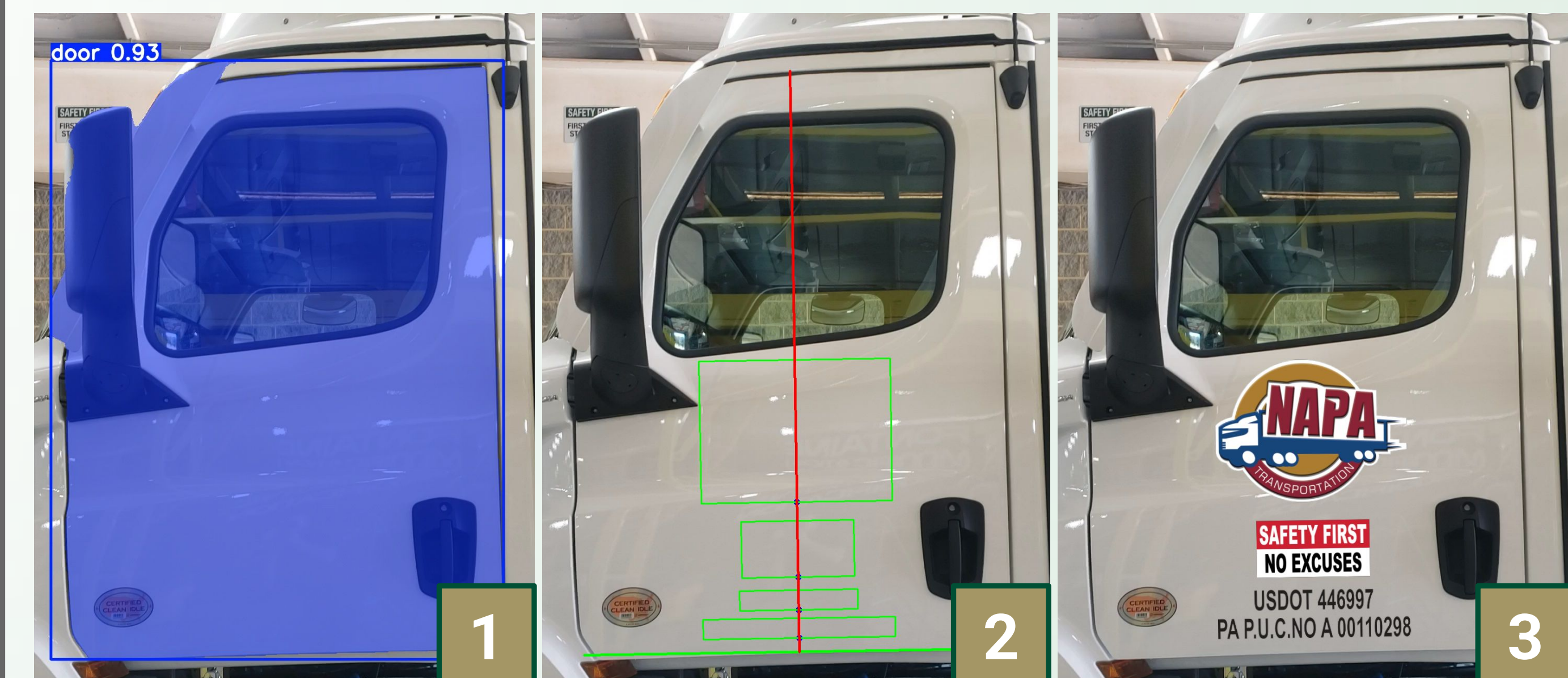
- Reduces process time by **>57%**
- Projection precision of **$\pm 0.25"$**
- Improves operator **confidence**
- Flexible for multiple **decal jobs**
- Visible in **bright shop environment**
- Operating temperature: **32°–105°F**
- Mobile design w/ **4 locking casters**
 - Push handle and **workstation**
- Compact design for **wall/ceiling mount**
 - **13.5 x 8 x 10 inches** (sensing head)
- **>6,200 lines of code**

Design Overview



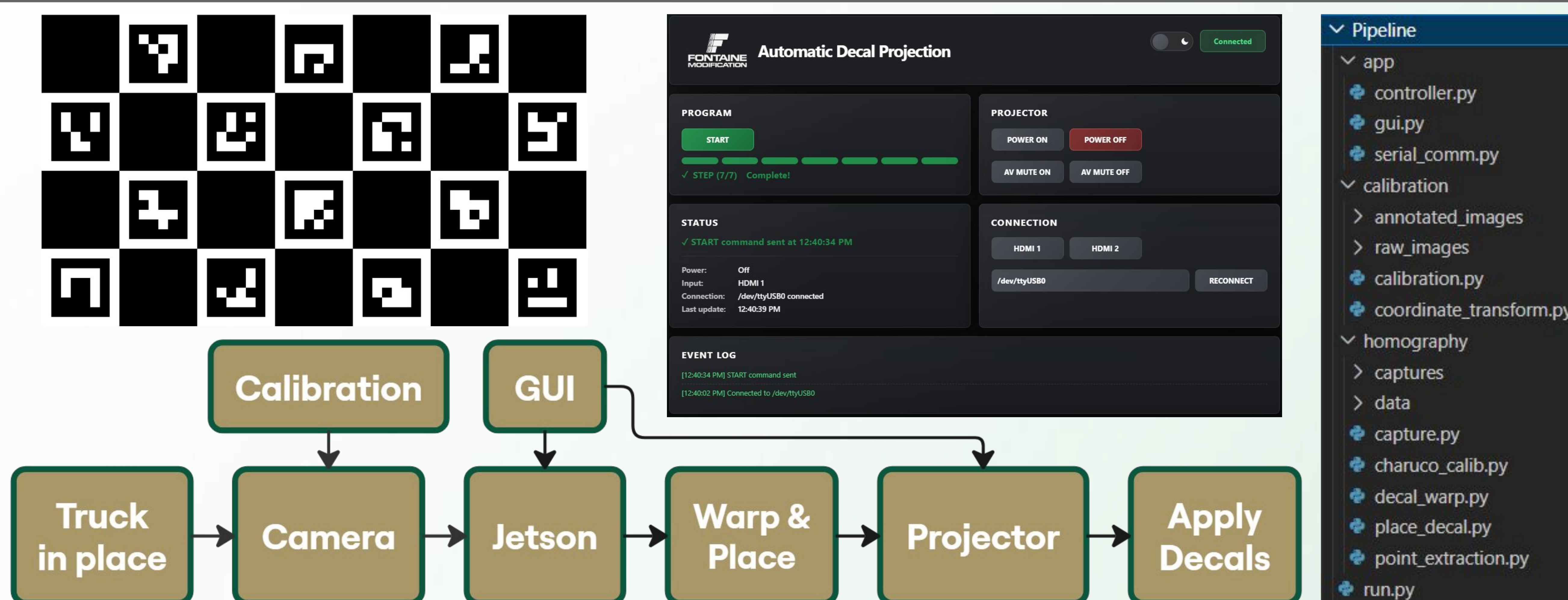
Testing & Impact

- **[1]** Testing reliability and accuracy of door detection with custom trained YOLOv8 segmentation model
- **[2]** Verification of contour extraction and homography-based warping from ChArUco calibration
- **[3]** Analyzing digital decal placement on the truck door



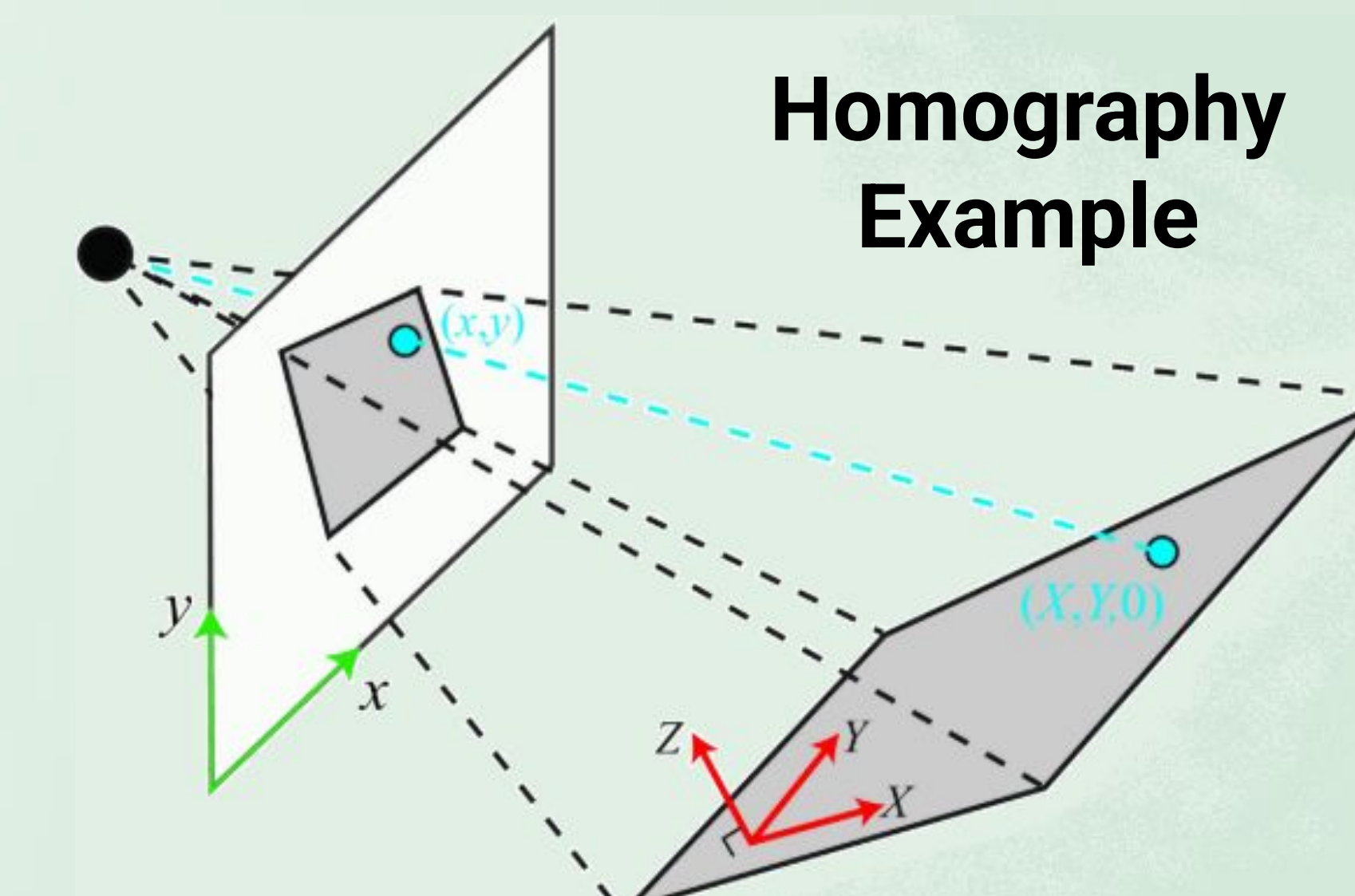
- Streamlined workflow with operator-friendly GUI for confident decal placements with minimal training
- System fully autonomous from end-to-end, image capture to digital decal projection

System Overview



Software

- **Python** – Programming Language
- **OpenCV** – Computer Vision Library
- **YOLOv8** – Segmentation Model
- **ChArUco** – Door Pose Mapping
- **RS-232** – Projector Communication
- **FastAPI & Uvicorn** – GUI & Web Server



Budget

